

Andrea Goh

☎ 317-363-3305 ✉ andreagoh0101@gmail.com 🔗 linkedin.com/in/goh-andrea 📍 Lafayette, IN 47905

Education

Purdue University West Lafayette, IN
Masters in Computer Engineering | Specialization: AI/ML | GPA: 3.57/4.00 08/24 - 05/26
BS in Aeronautical and Astronautical Engineering | Specialization: Controls | GPA: 3.77/4.00 08/20 - 05/24

- Minor in Mathematics
- Dean's List 8/8 semesters

Technical Skills

Languages: Python, C/C++, Go, SQL, MATLAB
AI/ML: PyTorch, TensorFlow, OpenCV, TensorBoard, Scikit-learn, NLP, LLMs, Generative AI, Deep Learning, Computer Vision, Transformers, CNNs, Diffusion Models, SLAM
AI Engineering: Prompt Engineering, LLM Evaluation, Model Benchmarking, AI Output Scoring, RAG, AWS Bedrock
Tools & Technologies: AWS, React, Docker, Git, Linux, HTML, CSS

Experience

ArgonDigital Austin, TX | Remote
AI Development Intern 07/26 – Present

- **Chat-with-AI feature:** Developing a chat-with-AI feature using RAG with contextual retrieval on Amazon Bedrock, enabling users to query documents/project data with grounded, context-aware responses.
- **LLM-based requirements generation:** Improved LLM-based requirements generation via prompt engineering, evaluating multiple foundation models via AWS Bedrock on quality, consistency, latency, and cost.
- **Evaluation rubric:** Designed an evaluation rubric for AI-generated requirements, combining deterministic programmatic checks with LLM-based scoring of subjective quality criteria.
- **AI Output Scoring System:** Generalized the rubric into a configurable AI Output Scoring System usable across multiple use cases beyond requirements generation, with admin-only visibility and configuration-driven criteria that update without code changes or deployments.
- Architected the scoring system to run asynchronously without impacting AI generation latency, supporting data-driven benchmarking of model and prompt configurations.
- **Benchmark dataset:** Built a benchmark dataset from synthetically generated transcripts derived from past project requirements, enabling systematic and reproducible evaluation.
- **ArgonSense enhancements:** Developed product enhancements and bug fixes for ArgonSense, including model description tooltips and LLM-generated feedback titles to improve usability and information clarity.

Purdue Robotic Vision Lab West Lafayette, IN
Computer Vision Research Assistant (RA) 01/25 – 05/26

- Built a custom tokenizer to encode OpenStreetMap data for a multimodal transformer designed to generate unseen data in a target modality conditioned on sparse, incomplete modalities, filling gaps across timelines and sensor types.
- Designed automated data ingestion pipelines (APIs, web scraping) with Python-based cleaning, transformation, and validation to construct the custom OSM dataset used to train the tokenizer.
- Designed and executed controlled experiments on the multimodal transformer, using quantitative metrics to guide tokenizer and model selection for production deployment.
- Performed quantitative analysis on multimodal sensor data (LiDAR, RGB, IMU) to improve the robustness of multimodal transformers under real-world distribution shifts, for self-driving car applications.
- Engineered custom data loaders and vectorization pipelines for computer vision projects, implementing YOLO-style vectorization for object detection and supporting label injection for image segmentation and classification, reducing training bottlenecks.
- Developed a 3D stereo reconstruction pipeline from scratch using classical vision theory and plain mathematics (no OpenCV or other CV libraries used).
- Built and trained deep learning models (PyTorch, TensorFlow) for generative AI, image segmentation, and classification tasks.

Purdue University

Graduate Teaching Assistant (GTA)

West Lafayette, IN

08/24 – 05/26

- Head GTA, Advanced C Programming: Led a 20-member instructional team supporting 469 students; coordinated course logistics and contributed to the creation of 25+ assignments and 3 exams.
- Head GTA, Python for Data Science: Designed exams and grading scripts, coordinated course operations, and taught advanced data science and ML concepts during office hours.
- GTA, Computer Security: Mentored students in debugging C/Python programs and cryptographic implementations, reinforcing secure coding best practices.

Purdue TNT Lab | Samsung

Research Assistant

West Lafayette, IN

05/24 – 08/24

- Co-authored manuscript, *Ultra-Small Damage Detection via EIT in Soft Nanocomposite Sensors*, submitted to the *IEEE Sensors Journal*; preprint available on *engrXiv*.
- Pioneered the integration of machine learning with electrical impedance tomography (EIT), driving the lab's adoption of ML-based approaches for damage localization.
- Designed a 3D-printed electrode prototype incorporating pogo pins as a novel electrode interface and integrated the experimental setup with Python for automated data collection and processing.
- Automated a simulation-to-training pipeline generating 4,500+ finite element method (FEM) datasets, accelerating ML model training by 3×.
- Developed ML-accelerated EIT with cross-functional team members, achieving 6× faster anomaly localization.
- Developed data preprocessing and model training workflows to transform FEM simulation outputs into datasets suitable for deep learning-based damage localization.

Thales

Modelling and Simulations Engineer Intern

Singapore

05/23 – 08/23

- Reduced COMSOL simulation runtime by 82% while maintaining over 99% accuracy through simulation and model optimization.
- Conducted parametric studies to optimize tunable piezoelectric lattice models for low-frequency hydrophone applications.
- Designed tunable lattice piezoelectric CAD models for purpose-built sonar and hydrophone applications and delivered a functional prototype.
- Developed a standardized workflow for simulation setup, meshing, and parameterization to support multi-application testing.
- Researched existing piezoelectric lattice architectures and evaluated design approaches to identify opportunities for improved hydrophone performance.

Projects

MedAR, Catapult Hacks 2025: AI x Startup Hackathon | Runner-up for Best AI-Driven Solution

- Integrated multiple ARKit and RealityKit Swift projects to build an AR surgical training prototype featuring hand-tracked tumor visualization and voice-command tool selection.
- Implemented dynamic 3D object placement, labeling, and interaction to support intuitive visualization of anatomical structures in augmented reality.
- Integrated SAM-based image segmentation and stereo vision for real-time hand tracking, achieving less than 5 mm tracking error on live hand gestures.

Financial Website Development | HTML, CSS, Java, Go, SQL

- Developed a responsive financial web application with user authentication, secure form validation, and account management functionality.
- Implemented a Go backend and SQL database integration to securely manage user credentials and account data.
- Designed REST-compliant API responses and standardized HTTP status handling for reliable client-server communication.
- Implemented responsive UI components and cross-browser compatibility across desktop and mobile interfaces.

Blimp Control | MATLAB, Simulink

- Developed a simulation of an aerial pursuit-evasion scenario incorporating trajectory planning and path optimization strategies.
- Designed and tuned PID controllers for forward velocity, altitude, and heading control of an autonomous blimp using Simulink.
- Evaluated controller performance through simulation-based testing under varying trajectories and operating conditions.